

Short Report: AO Unit Step-Size Calibration (2026 Repeat)

13 July 2026 -- COM6, automated MaxIm DL readout

Summary

Automated repeat of the 2024 AO step-size calibration: the notebook drove the AO over COM6, triggered MaxIm DL exposures via COM, and read centroids back automatically. Total run time ~14.5 min (13:04:10-13:18:37): Phase 1 (1 step) ~3.5 min, then Phase 2 (100 steps) and Phase 3 (1000 steps) each ~4.5+ min including a 60 s inter-phase cooldown and manual centroid re-seed. Setup: ZWO ASI6200MM-PRO (3.76 um/px), cooler -15 degC, 10 s exposures, commands GW/GN/GS/GT/K unchanged from 2024.

Headline: at 100 and 1000 steps, today's per-step movement is far more self-consistent (~0.019-0.021 um/step at both scales, vs. the 2024 report's ~10x drop between 100 and 1000 steps) and the four directions are close to orthogonal (88.9-91.6 deg between adjacent axes) and close to antiparallel in pairs (0.6-5.7 deg off 180 deg) -- see Section 2. Absolute step size is 2-20x smaller than the 2024 "New AO" figures (Section 3), which needs confirming as a hardware/config change before being reported as a precision gain.

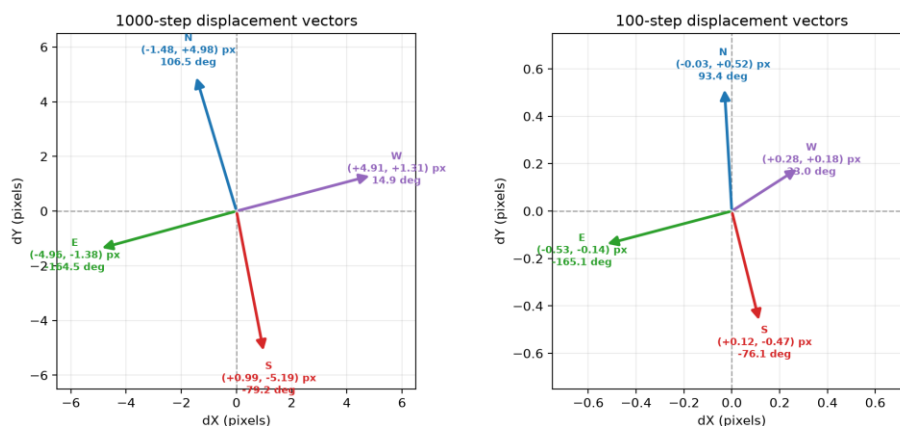
1. Results -- 13 July 2026

Phase	Dir	dX (px)	dY (px)	Dist (px)	Dist (um)	um/step	px/step
1 step	N	-0.046	0.003	0.046	0.173	0.1731	0.04600
1 step	S	-0.050	-0.059	0.077	0.290	0.2903	0.07700
1 step	E	0.010	0.063	0.064	0.241	0.2409	0.06400
1 step	W	-0.188	0.015	0.189	0.711	0.7110	0.18900
100 steps	N	-0.031	0.523	0.524	1.970	0.0197	0.00524
100 steps	S	0.116	-0.470	0.484	1.820	0.0182	0.00484
100 steps	E	-0.531	-0.141	0.549	2.065	0.0206	0.00549
100 steps	W	0.282	0.183	0.336	1.264	0.0126	0.00336
1000 steps	N	-1.479	4.978	5.193	19.530	0.0195	0.00519
1000 steps	S	0.992	-5.190	5.284	19.870	0.0199	0.00528
1000 steps	E	-4.965	-1.375	5.152	19.370	0.0194	0.00515
1000 steps	W	4.913	1.309	5.084	19.120	0.0191	0.00508

2. Movement Direction -- Is It Orthogonal?

Centre-to-after-move displacement vectors for each direction, both step counts. Angle measured from the +X pixel axis (atan2(dY,dX)).

AO displacement vectors -- centre to after-move (13 July 2026)



Axis pair	Ideal (deg)	1000-step (deg)	1000-step dev.	100-step (deg)	100-step dev.
N vs S	180	174.3	5.7	169.5	10.5
E vs W	180	179.4	0.6	161.9	18.1
N vs E	90	88.9	1.1	101.5	11.5
N vs W	90	91.6	1.6	60.4	29.6

At 1000 steps the four directions are close to orthogonal/antiparallel (0.6-5.7 deg deviation). At 100 steps the deviation is much larger (10.5-29.6 deg) because the displacement itself (0.3-0.5 px) is close to the centroid measurement noise floor -- same caveat as the 1-step data, just less extreme. The 1000-step result is the more trustworthy orthogonality check.

3. Comparison Against the 2024 Report ("New AO")

Steps	2024 report (um/step)	Today (um/step)	Ratio
100 steps	0.310 - 0.390	0.0126 - 0.0206	~17-20x smaller
1000 steps	0.0341 - 0.0398	0.0191 - 0.0199	~1.8-2x smaller

4. Key Findings

- 1-step results are noise-dominated (0.05-0.19 px, same order as centroid scatter) – will not use for per-step precision claims.
- 100- and 1000-step results agree closely (0.0126-0.0206 vs 0.0191-0.0199 um/step) -- more linear than 2024's ~10x drop.
- 1000-step directions are close to orthogonal/antiparallel (Section 2); 100-step angles are noisier for the same reason as the 1-step data.
- Absolute step size is 2-20x smaller than 2024's "New AO" -- confirm same physical unit/firmware before reporting as an improvement.

Appendix A -- Calculations

Displacement (px) = $\sqrt{(X_{\text{after}} - X_{\text{before}})^2 + (Y_{\text{after}} - Y_{\text{before}})^2}$ Displacement (um) = Displacement (px) x 3.76 um/px

um/step = Displacement (um) / steps px/step = Displacement (px) / steps angle (deg) = $\text{atan2}(dY, dX) \times 180/\pi$

Example (1000-step N): dX=-1.479 px, dY=4.978 px -> dist = $\sqrt{1.479^2 + 4.978^2}$ = 5.193 px = 19.53 um -> 0.01953 um/step = 0.00519 px/step -> angle = $\text{atan2}(4.978, -1.479)$ = 106.5 deg.

Source: AO_COM6_auto_results_20260713.xls (12 rows).

Reference: reference_AO_report_2024.pdf.

I can share the actual csv if needed where I have done these above calculations for reference ~ Biswajit